COMPLETION OF A TOPOLOGICAL MODULE WITH AN I-ADIC OPEN SUBMODULE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one surjection between completions; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The completion of a module with respect to a finitely generated ideal, and the topology on it. Algebra, Lemma algebra-lemma-hathat-finitely-generated, invoked at proof lines 1036-1039. Gives that the completion of the chosen open submodule has I-adic topology.
- Exactness of completions for a short exact sequence of topological modules in this setting. Same chapter, Lemma lemma-ses, invoked at proof lines 1046-1048. Produces the surjectivity of the induced map on completions of the finite direct sum.
- Finitely generated ideals have finitely generated powers. Standard, used at proof lines 1040-1043. Supplies the generators whose map is continuous and open.

Source reference. Original lemma 05GG: sources/stacks-chow/tags.txt, source lines 7060–7060. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AS0: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 998–998. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let R be a topological ring. Let M be a topological R-module. Let $I \subset R$ be a finitely generated ideal. Assume M has an open submodule whose topology is I-adic. Then $M^\wedge$ has an open submodule whose topology is I-adic and we have $M^\wedge/I^nM^\wedge = M/I^nM$ for all $n \geq 1$.*

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Proof. Let $M' \subset M$ be an open submodule whose topology is I-adic. Then $\{I^n M'\}_{n \geq 1}$ is a fundamental system of open submodules of M. Thus $M^\wedge = \lim M/I^n M'$ contains $(M')^\wedge = \lim M'/I^n M'$ as an open submodule and the topology on $(M')^\wedge$ is I-adic by Algebra, Lemma 05GG. Since I is finitely generated, I^n is finitely generated, say by $f_1, \ldots, f_r$. Observe that the surjection $(f_1, \ldots, f_r) : M^{\oplus r} \to I^n M$ is continuous and open by our description of the topology on M above. By Lemma 0AS0 applied to this surjection and to the short exact sequence $0 \to I^n M \to M \to M/I^n M \to 0$ we conclude that

$$(f_1, \ldots, f_r) : (M^\wedge)^{\oplus r} \longrightarrow M^\wedge$$

surjects onto the kernel of the surjection $M^\wedge \to M/I^n M$. Since $f_1, \ldots, f_r$ generate I^n we conclude. $\square$